Problem Set 7

1) Consider a theory of real scalar fields A, B, C with Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial^\mu A \partial_\mu A - \frac{1}{2}m_A^2 A^2 - \frac{1}{2}\partial^\mu B \partial_\mu B - \frac{1}{2}m_B^2 B^2 - \frac{1}{2}\partial^\mu C \partial_\mu C - \frac{1}{2}m_C^2 C^2 + g\,ABC$$

Write down the tree level amplitudes for these processes

$AA \to AA$ $AA \to BC$

$AA \to AB$ $AB \to AB$

$AA \to BB$ $AB \to AC$

2) Rutherfor Scattering

The cross section for scattering an electron by the Coulomb field of a nucleus can be computed, to lowest order, without quantizing the electromagnetic field.

We treat the field $A_\mu(x)$ as a classical potential & consider the interaction

$$\hat{H}_I = \int d^3x \; e\hat{\bar{\Psi}}\gamma^\mu\hat{\Psi} A_\mu$$

where $\psi(x)$ is a quantized Dirac field.

a) Compute the T-matrix for electron scattering off a localized classical potential to the lowest order.

b) If $A_\mu(x)$ is time independent it is natural to define

$$\langle p'|iT|p\rangle \equiv i\mathcal{M}\cdot(2\pi)\,\delta(\overbrace{E_f-E_i}^{\text{final \& initial energies}})$$

Show that the cross section for scattering off a time-independent localized potential is

$$d\sigma = \frac{1}{v_i}\frac{1}{2E_i}\frac{d^3p_f}{(2\pi)^3}\frac{1}{2E_f}|M(p_i - p_f)|^2 \, 2\pi\,\delta(E_f - E_i)$$

v_i: initial velocity

Integrate over $|p_f|$ to find $\frac{d\sigma}{d\Omega}$.

3) Any particle that couples to electron can produce a correction to $g-2$.
Because the $g-2$ agrees with QED to high accuracy, the corrections to $g-2$ constrain properties of hypothetical particles.

a) Consider the Higgs boson h which couples to electron as

$$H_{int} = \int d^3x \, \frac{\lambda}{\sqrt{2}} h \bar{\psi} \psi$$

Compute the contribution of a virtual Higgs boson to electron's $g-2$, in terms of λ and the m_h of Higgs.

b) If the experimental value of $a = g-2$ agrees with a_{QED} up to 10^{-10}. What limit does this place on λ & m_h?